

# YU YUAN

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## RESEARCH INTERESTS

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Physics-Consistent and Controllable Video Generation, Physical World Models, Generative Models, Multimodal AI, Agentic AI, 3D/4D Generation, Computational Photography, Image/Video Editing/Enhancement, VLM.

## EDUCATION

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### Purdue University

PhD Student in Electrical and Computer Engineering    Advisor: Stanley H. Chan

West Lafayette, IN, US

08/2023 – 12/2026 (expected)

### Shanghai Jiao Tong University

MS in Aeronautical and Astronautical Science and Technology

Shanghai, China

09/2020 – 06/2023

### Shanghai Jiao Tong University

BEng in Aeronautics and Astronautics Engineering

Shanghai, China

09/2016 – 06/2020

Minor Degree in Administration Management

10/2018 – 06/2020

## EXPERIENCE

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### Adobe Research

Research Intern (Multimodal Video Foundation Models Team)

San Jose, CA

05/2026 – 08/2026

- **Led the 0-to-1 development of a novel causal video dynamics editing framework**, driving the full stack from problem formulation and large-scale data generation to video foundation model training, evaluation, and system integration.
- **Built a scalable multimodal data engine** that automatically generated paired counterfactual videos through data sourcing, filtering, VLM-based edit planning, video generation, quality control, and benchmark construction.
- **Delivered the first general-purpose framework for scene dynamics editing** while preserving scene consistency, resulting in 1 research paper submission and 2 patent filings.

### Purdue University

Research Assistant (Intelligent Imaging Lab)

West Lafayette, IN

08/2023 – Present

- **Led multiple end-to-end research and engineering projects** in video generation, world modeling, physics-aware generative AI, and computational photography, spanning problem formulation, model development, large-scale training, evaluation, and open-source release.
- **OptiWorld – Optimal-Controlled Video World Generation**: Introduced optimal control into video world generation, jointly optimizing object dynamics under physical, safety, efficiency, and smoothness constraints for controllable trajectory generation.
- **SeeU – 4D Understanding and Video Generation**: Developed a unified continuous 4D world model that learns spatiotemporal dynamics from 2D observations, enabling generation and editing at unseen times and viewpoints.
- **NewtonGen – Physics-Consistent and Controllable Video Generation**: Developed a physics-consistent and controllable video generation framework with Neural Newtonian Dynamics, unifying diverse physical dynamics through neural ODEs.
- **Generative Photography – Camera-Controlled Image/Video Generation**: Introduced a camera-physics-aware generative photography paradigm for scene-consistent control over aperture, focal length, shutter speed, and color temperature.
- **Authored 4 first-author papers** at top-tier venues including ICLR, CVPR, and NeurIPS, spanning video generation, world modeling, physical dynamics, and controllable generative models.

## SELECTED PUBLICATIONS

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**Yu Yuan**, Yawen Lu, Guoxian Song, Kevin Duarte, Ratheesh Kalarot, Di Chang, Xijun Wang, Stanley H. Chan.

***Reimagine Video Dynamics***

2026 [In Submission]

**Yu Yuan**, Jianhao Yuan, Xijun Wang, Daiqing Li, Liu He, Lu Ling, Stanley H. Chan.

***OptiWorld: Optimal Control for Video World Generation under Physical Constraints***

2026 [Under Review]

**Yu Yuan**, Tharindu Wickremasinghe, Zeeshan Nadir, Xijun Wang, Yiheng Chi, Stanley H. Chan.

***SeeU: Seeing the Unseen World via 4D Dynamics-aware Generation***

CVPR 2026

**Yu Yuan**, Xijun Wang, Tharindu Wickremasinghe, Zeeshan Nadir, Bole Ma, Stanley H. Chan.

***NewtonGen: Physics-Consistent and Controllable Text-to-Video Generation via Neural Newtonian Dynamics.***

ICLR 2026

**Yu Yuan**, Xijun Wang, Yichen Sheng, Prateek Chennuri, Xingguang Zhang, Stanley H. Chan.

***Generative Photography: Scene-Consistent Camera Control for Realistic Text-to-Image Synthesis.***

CVPR 2025 [Highlight & Demo]

**Yu Yuan**, Yiheng Chi, Xingguang Zhang, Stanley H. Chan.

***iHDR: Iterative HDR Imaging with Arbitrary Number of Exposures.***

ICIP 2025

Xingguang Zhang, Nicholas Chimitt, Xijun Wang, **Yu Yuan**, Stanley H. Chan.

***Learning Phase Distortion with Selective State Space Models for Video Turbulence Mitigation.***

CVPR 2025 [Highlight]

Joonyeoup Kim, **Yu Yuan**, Xingguang Zhang, Xijun Wang, Stanley H. Chan.

***Astrophotography Turbulence Mitigation via Generative Models.***

ICIP 2025

**Yu Yuan**, Jiaqi Wu, Lindong Wang, Zhongliang Jing, Henry Leung, Shuyuan Zhu, Han Pan.

***Learning to Kindle the Starlight.***

arXiv 2022

**Yu Yuan**, Jiaqi Wu, Zhongliang Jing, Henry Leung, Han Pan.

***Multimodal Image Fusion based on Hybrid CNN-Transformer and Non-local Cross-modal Attention.***

arXiv 2022

## PATENT

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Learning a Compact Dynamics Token for Precise Video Motion Reconstruction and Editing. [In Submission].

A Data Pipeline for Generating Counterfactual Video Pairs from Real Videos. [In Submission].

Trajectory resolving and registering method based on autonomous underwater robot inertial navigation and ultra-short baseline positioning sensor. CN 114993313 A.

## AWARDS & COMPETITIONS

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Ivy League Scholarship for Exceptional Students (Innovation)

Top 1%

Outstanding Graduate of Shanghai Jiao Tong University

Top 10%

Shanghai International Creators Competition: Special Competition of Drones

First Prize

DELL AI for Social Innovation Competition: AI/ADAS of Intelligent Car

National Second Place

## OTHER ACTIVITIES

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**Teaching Assistant**, Automatic Control Theory, Shanghai Jiao Tong University

**Workshop Organizer**, WACV GAIP 2026

**Reviewer**, CVPR, ICCV, ECCV, ICLR, NeurIPS, WACV, CVPRW, ICIP, ICASSP

**IEEE Graduate Student Member**

## SKILLS

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**Languages:** English (Professional Working Proficiency), Chinese (Native)

**Programming:** Python, PyTorch, HTML, JavaScript, Matlab, C/C++ (Basic)

**Technologies:** SFT, Post-training, LoRA Finetuneing, Diffusers, Transformers, 3DGS, CUDA, OpenCV, NumPy, Git, LaTeX, Linux, Slurm, Markdown, Gradio, NVIDIA Jetson, Adobe Photoshop, Adobe Premiere

**Others:** Photography, Fishing, Drones, Rowing